

CREATE • RUN • COMPARE • EXPLAIN

Charlottesville–Albemarle VisionEval Walkthrough

A screenshot-demonstrated, end-to-end scenario exercise for VisionEval Workbench 1.0.0

REGION	Charlottesville–Albemarle Metropolitan Planning Organization
EXERCISE	Baseline + three 2045 growth scenarios

USE THIS GUIDE Follow the numbered procedures in order. Each screenshot is paired with the evidence it confirms and the next action to take.

Prepared 17 August 2026 • Companion to the VisionEval Workbench wiki

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1. Before you begin

This tutorial assumes VisionEval Workbench is already installed.

Installation guides

- Mac Intel installation and setup
- Mac Silicon installation and setup
- Windows installation and setup

WIKI COVERAGE The wiki explains individual features. This companion adds the missing screenshot-driven, start-to-finish exercise using one real MPO, three scenarios, actual runs, and observed comparisons.

What you need

- VisionEval Workbench 1.0.0 and a functioning VisionEval runtime.
- The current Virginia MPO Regional Data package (demonstrated with version 2026.08.12.3).
- Enough disk space and time for four completed datastores: baseline, Faster Growth, Suburbanization, and Urbanization.

2. Exercise design

The exercise adapts the supplied PlanRVA planning concept to the Charlottesville–Albemarle MPO. The planning image is evidence for the scenario idea only; the table below is the operative tutorial specification.

Scenario	Geography	Population / housing	Employment	Method
Faster Growth	Charlottesville City + Albemarle County	+10%	+10%	Batch Change
Suburbanization	Population/housing: Albemarle County; employment: Charlottesville City	+10%	+10%	Single-file editor
Urbanization	Charlottesville City	+10%	+10%	Single-file editor

Files and columns changed

Input file	2045 columns increased by 10%
azone_gq_pop_by_age.csv	All six group-quarters age fields
azone_hh_pop_by_age.csv	All six household age fields
bzone_dwelling_units.csv	SFDU, MFDU, GQDU
bzone_employment.csv	TotEmp, RetEmp, SvcEmp
	ROUNDING RULE All edited outputs are whole numbers. Workbench multiplies the selected 2045 values by 1.10 (10%) and rounds to the nearest whole number. The 2024 rows remain unchanged.

3. Prepare a clean workspace

- 1. Close active projects and note the current Workbench workspace location so it can be restored after the tutorial.
- 2. Create a new empty folder named for the tutorial, then choose it as the Workbench workspace.
- 3. Confirm that Create shows no saved projects and Develop shows no installed regional package.

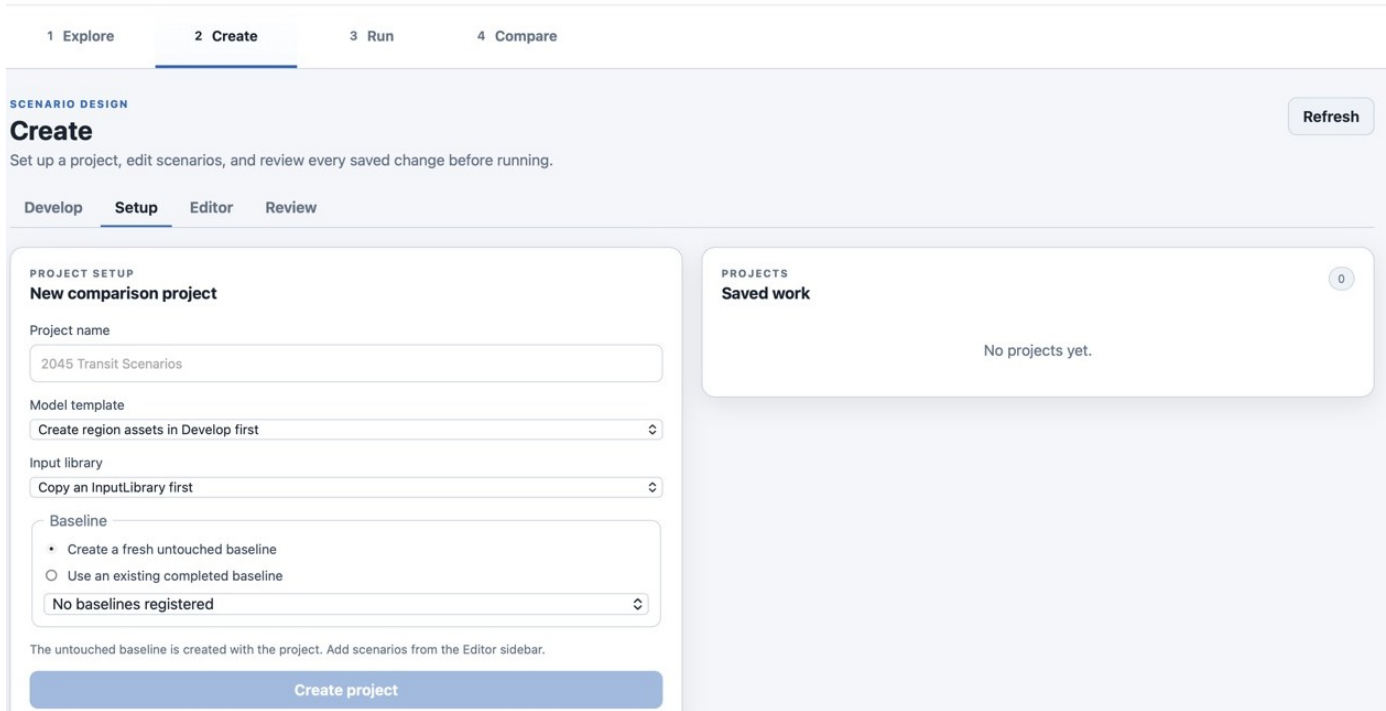


Figure 1. The clean tutorial workspace contains no projects or saved work.

4. Install and preview the regional package

4. Open Create → Develop. If no package is installed, select Install regional package.

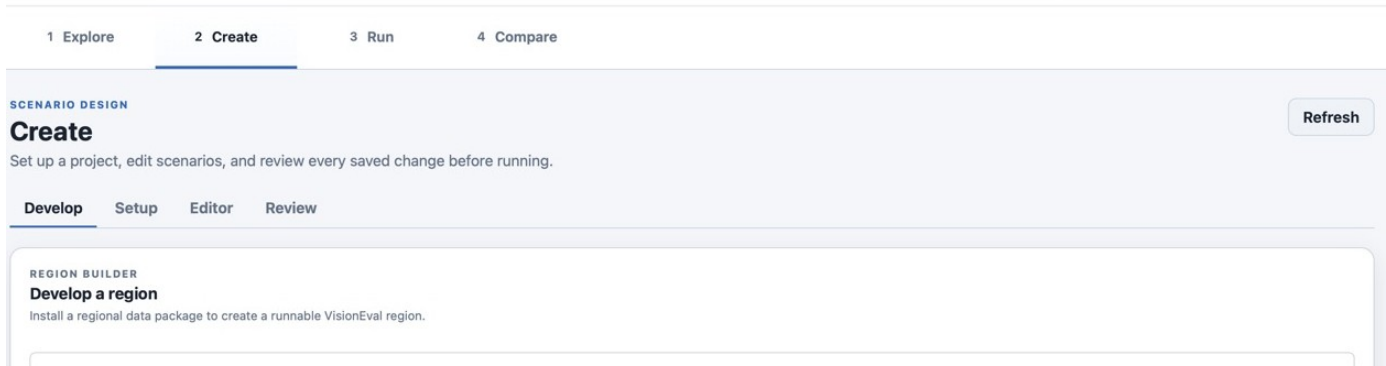


Figure 2. Develop identifies that a regional package is required before an MPO can be built.

5. Install the Virginia MPO Regional Data package and wait for Workbench to validate it.

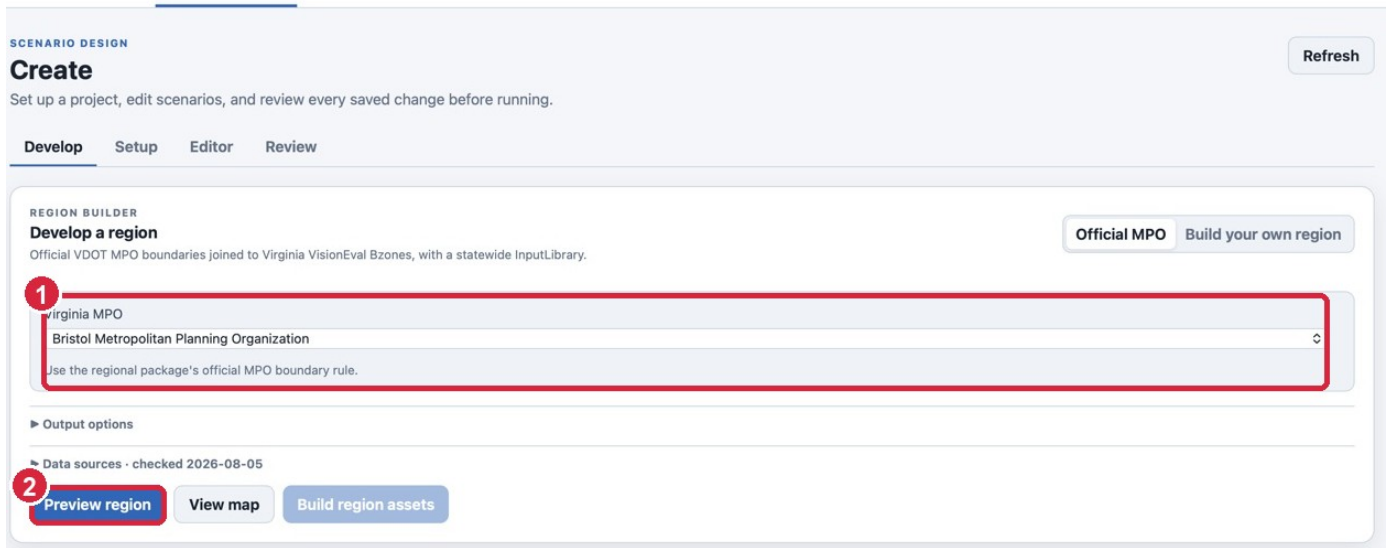


Figure 3. The validated Virginia MPO package is ready to use.

- 1 Package identity and version. 2 Preview region begins regional asset creation.

- Choose Charlottesville–Albemarle Metropolitan Planning Organization from the MPO selector.

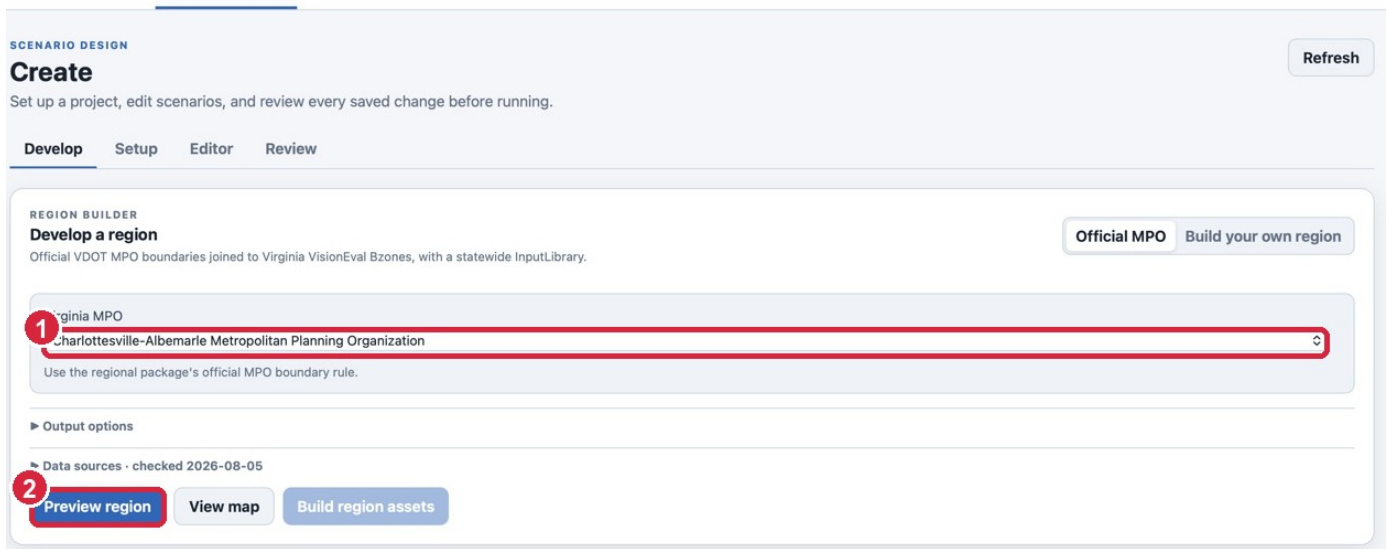


Figure 4. The selected MPO becomes the source region for the tutorial.

- Selected MPO. 2 Preview the regional boundary before building.
- Open View map and verify that the boundary and zone geography look plausible.

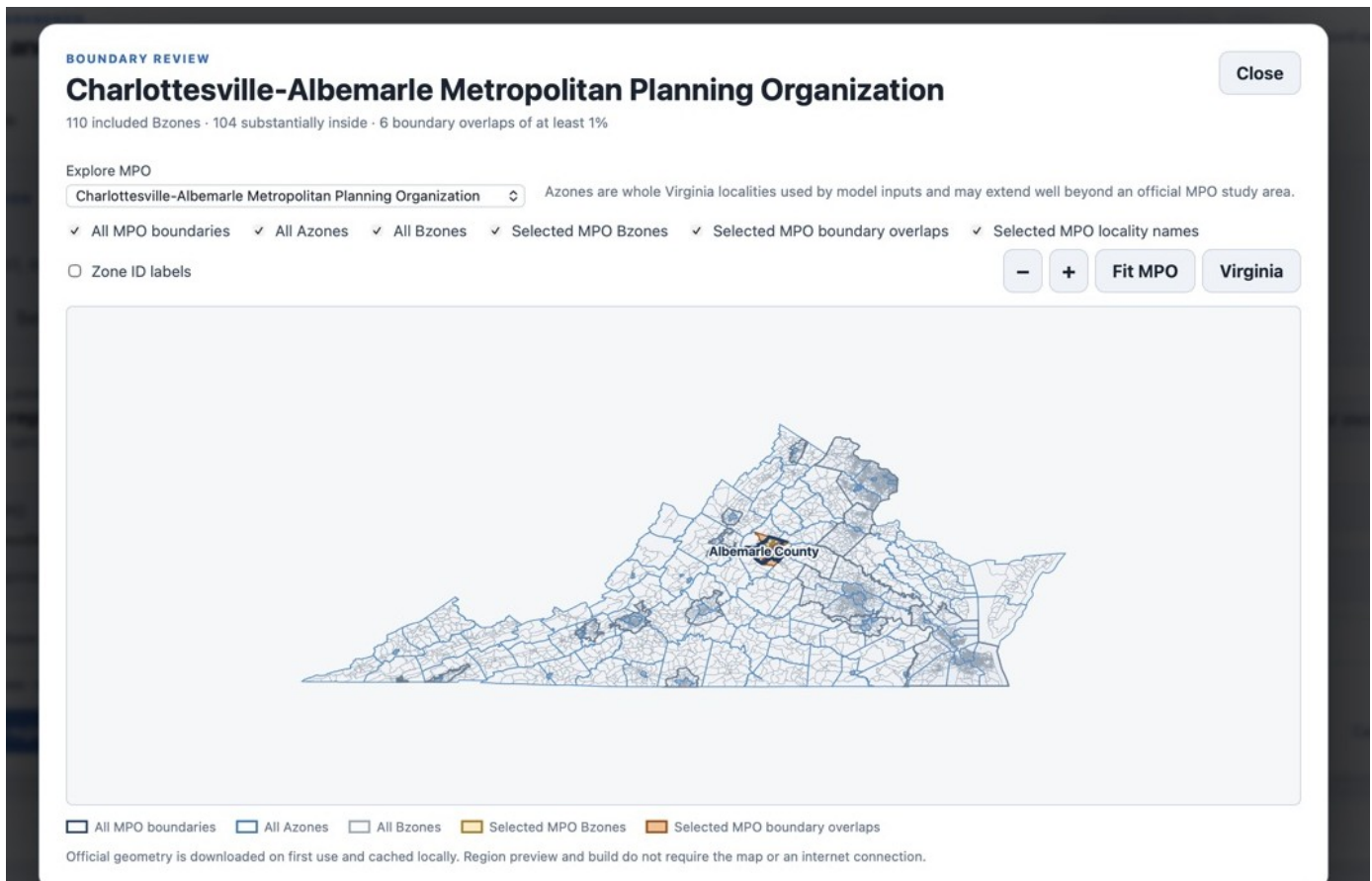


Figure 5. The preview map confirms the Charlottesville–Albemarle boundary and zone coverage.

8. Review the package summary before building the InputLibrary and model template.

REGION BUILDER
Develop a region
 Official VDOT MPO boundaries joined to Virginia VisionEval Bzones, with a statewide InputLibrary.

Official MPO | Build your own region

Virginia MPO
 Charlottesville-Albemarle Metropolitan Planning Organization

Use the regional package's official MPO boundary rule.

► Output options

► Data sources · checked 2026-08-05

Preview region | View map | **Build region assets**

Cached locally

1 **REGION PREVIEW**
Charlottesville-Albemarle Metropolitan Planning Organization
 Prepared from Virginia InputLibrary

110 included Bzones	2 Azones / localities	6 Boundary overlaps included
-------------------------------	---------------------------------	--

2 **included localities:** Albemarle County, Charlottesville City

Figure 6. The region preview confirms 110 included Bzones, two localities, six boundary overlaps, and 52 source inputs.

1 Selected MPO identity. 2 Preview counts and locality coverage. 3 Build region assets only after the preview is correct.

5. Build assets and create the project

9. Select Build region assets. Wait for both the InputLibrary and model template to finish.

SCENARIO DESIGN

Create Refresh

Set up a project, edit scenarios, and review every saved change before running.

Develop **Setup** Editor Review

PROJECT SETUP

New comparison project

Built region selected: Charlottesville-Albemarle Metropolitan Planning Organization

Project name
Charlottesville-Albemarle Metropolitan Planning Organization

Model template
Charlottesville-Albemarle Metropolitan Planning Organization

Input library
Charlottesville-Albemarle Metropolitan Planning Organization - 51 CSVs

Baseline
☒ Create a fresh untouched baseline
☐ Use an existing completed baseline
 No baselines registered

PROJECTS 0

Saved work

No projects yet.

Figure 7. The built region assets are available to the project setup form.

10. Enter the project name Charlottesville-Albemarle Growth Demonstration and keep the built regional assets selected.

SCENARIO DESIGN

Create Refresh

Set up a project, edit scenarios, and review every saved change before running.

Develop **Setup** Editor Review

PROJECT SETUP

New comparison project

Built region selected: Charlottesville-Albemarle Metropolitan Planning Organization

1 Project name
Charlottesville-Albemarle Growth Demonstration

Model template
Charlottesville-Albemarle Metropolitan Planning Organization

Input library
Charlottesville-Albemarle Metropolitan Planning Organization - 51 CSVs

2 Baseline
☒ Create a fresh untouched baseline
☐ Use an existing completed baseline
 No baselines registered

PROJECTS 0

Saved work

No projects yet.

Figure 8. The project setup is complete and ready to create.

- 1 Exact project name. 2 Create the project after checking the regional assets.

11. Create the project and confirm the baseline appears in the scenario editor.

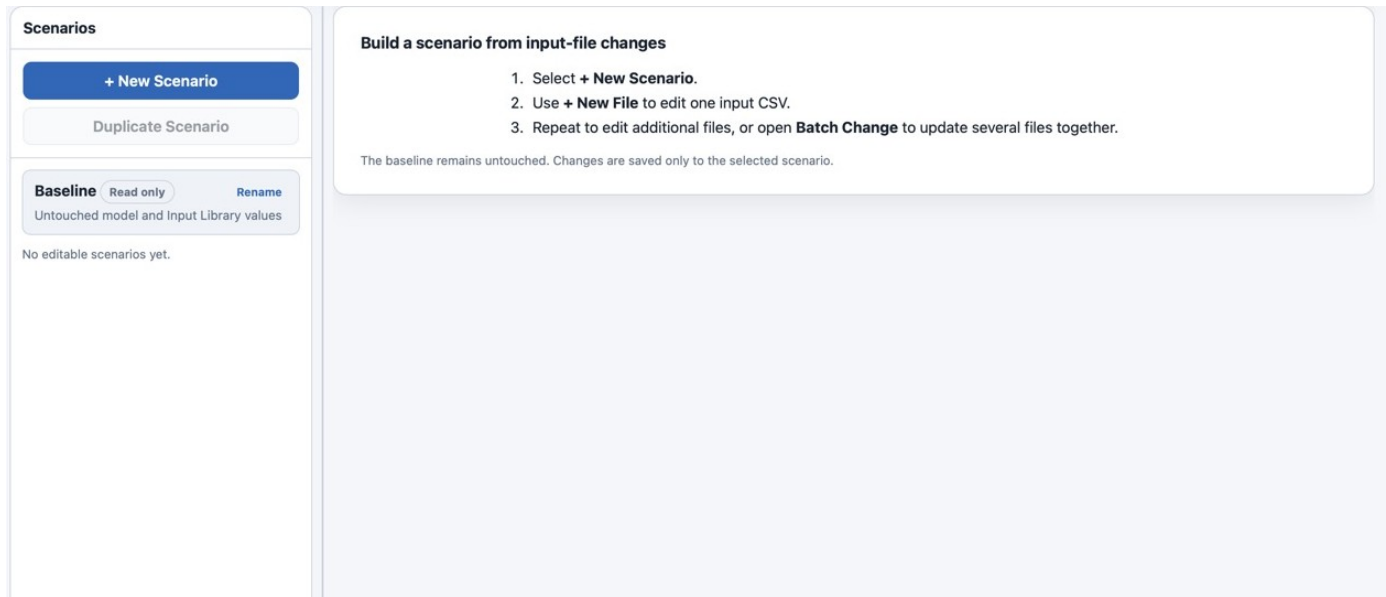


Figure 9. The new project opens with a baseline and no scenario overlays.

6. Create the three scenarios

6.1 Faster Growth — Batch Change

12. Create a scenario named Faster Growth and add a note that all 2045 population, housing, and employment inputs increase by 10% in both localities.

Scenarios

+ New Scenario

Duplicate Scenario

Baseline Read only Rename
Untouched model and Input Library values

Faster Growth > ✕

+ New File

EDIT ONE INPUT FILE

Faster Growth Save File Changes **WS**

Apply Preview changes the temporary working copy. Save File Changes persists this file and its note to the scenario.

Saved

Input file
Choose input file

Filters and change function
Filter locations, choose a year and columns, then preview a repeatable change.

Location type Location search Target year Operation Value

Choose input file

Locations ☐ Select all locations Columns ☐ Select all columns Clear

Choose a location type.

Choose an input file.

Apply Preview Revert Preview Undo Redo

Figure 10. Faster Growth is a separate scenario overlay; the baseline remains read-only.

- Open Batch Change and select the four input files. Select the listed columns, year 2045, both localities, Increase by %, and value 10.

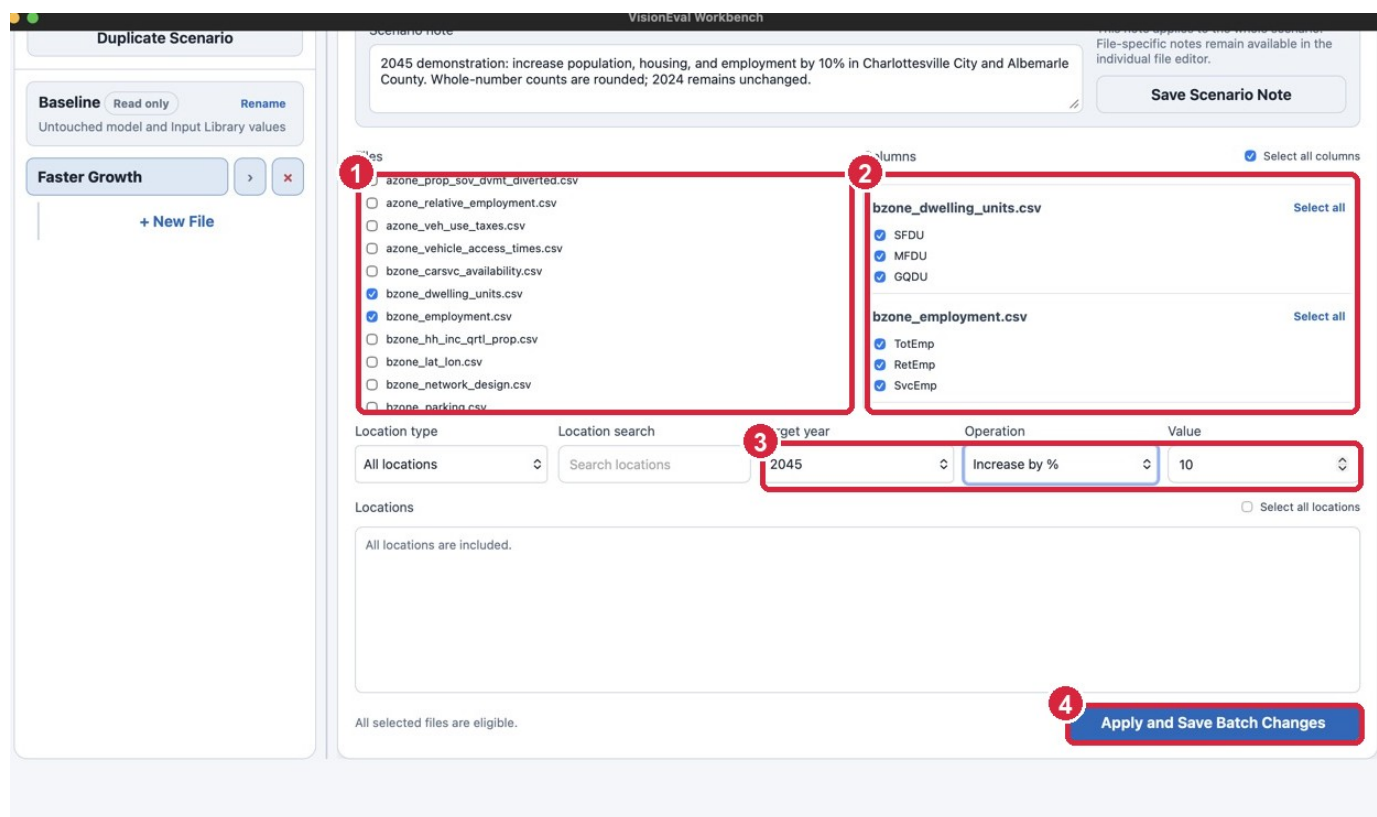


Figure 11. Batch Change applies one consistent rule across the Faster Growth input set.

- 1 Four target files. 2 Required population, housing, and employment columns. 3 Target year and percentage operation. 4 Apply and save only after every selector is verified.

14. Apply the preview, inspect the changed values, and save all four files to Faster Growth.

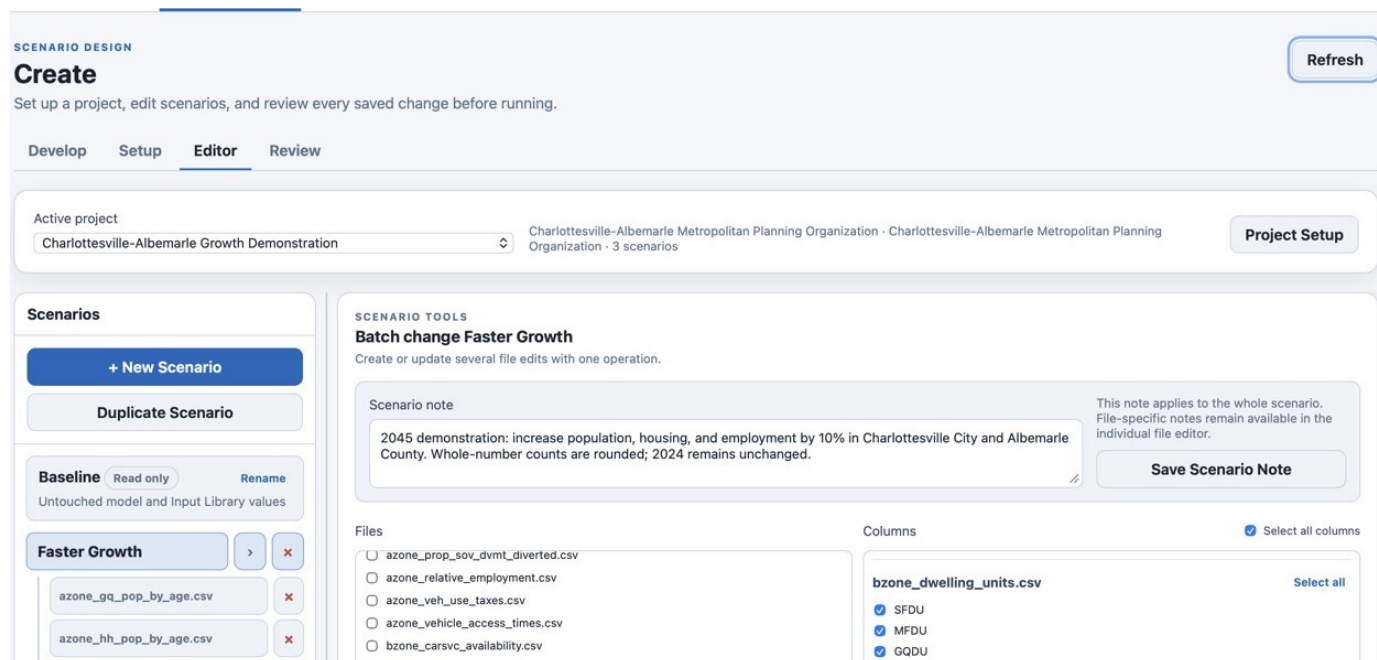


Figure 12. All three scenarios exist and the expected files are saved as overlays.

6.2 Suburbanization — Single-file editor

15. Create Suburbanization. In the two population files and dwelling-units file, filter to Albemarle County and 2045, then increase the required columns by 10%.

BaselineRead onlyRename
Untouched model and Input Library values

Faster Growth

azone_gq_pop_by_age.csv

azone_hh_pop_by_age.csv

bzone_dwelling_units.csv

bzone_employment.csv

+ New File

1

Suburbanization

azone_gq_pop_by_age.csv

azone_hh_pop_by_age.csv

bzone_dwelling_units.csv

bzone_employment.csv

+ New File

Urbanization

azone_gq_pop_by_age.csv

Filters and change function

Filter locations, choose a year and columns, then preview a repeatable change.

Location type

Location search

Target year

Operation

Value

County

Type to filter locations

2045

Set to

Locations

Select all locations

Columns

Select all columns

Clear

Albemarle County

Charlottesville City

Age0to14

Age15to19

Age20to29

Age55to64

Age30to54

Age65Plus

Apply Preview

Revert Preview

Undo

Redo

Apply Preview changes only this temporary working copy. Save File Changes makes the preview and file note part of Review and the next run.

Saved change from baseline

Unsaved preview

3

2

Geo	Year	Age0to14	Age15to19	Age20to29	Age30to54	Age55to64	Age65Plus
Albemarle County	2024	18444	10279	14208	34449	14070	23469
Charlottesville City	2024	3458	2804	11254	14447	4356	6118
Albemarle County	2045	24683	12362	17019	46824	20373	31430
Charlottesville City	2045	7857	3066	12255	17851	5734	7448

Figure 13. The single-file preview isolates Albemarle County in 2045; the displayed 2024 row remains unchanged.

1 Suburbanization scenario and file. 2 The tan preview row is the edited 2045 record. 3 The scenario note states the geography and unchanged base year.

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16. In bzone_employment.csv, filter to Charlottesville City and 2045, then increase TotEmp, RetEmp, and SvcEmp by 10%.

Filters and change function
Filter locations, choose a year and columns, then preview a repeatable change.

Location type: County, Location search: Type to filter locations, Target year: 2045, Operation: Set to, Value: [dropdown]

Locations: ☐ Albemarle County, ☒ Charlottesville City

Columns: ☐ TotEmp, ☐ RetEmp, ☒ SvcEmp

Buttons: Apply Preview, Revert Preview, Undo, Redo

Apply Preview changes only this temporary working copy. Save File Changes makes the preview and file note part of Review and the next run.

Legend: ■ Saved change from baseline, ■ Unsaved preview

Geo	Year	TotEmp	RetEmp	SvcEmp
510030101001	2024	177	0	52
510030102011	2024	543	4	58
510030102012	2024	999	62	624
510030102013	2024	87	0	17
510030102021	2024	115	4	58
510030102022	2024	273	0	109
510030103011	2024	2962	381	1406
510030103021	2024	119	0	74

Figure 14. The employment editor uses the required fields and retains the 2024 records.

Notes
Track assumptions and why this file changes.

2045 only: increase TotEmp, RetEmp, SvcEmp by 10% for Charlottesville City. Whole-number counts are rounded; 2024 is unchanged.

Figure 15. The saved note records the different employment geography: Charlottesville City.

6.3 Urbanization — Single-file editor

17. Create Urbanization. Apply the four-file edit only to Charlottesville City in 2045.

+ New File

Suburbanization

azone_gq_pop_by_age.csv

azone_hh_pop_by_age.csv

bzone_dwelling_units.csv

bzone_employment.csv

+ New File

1Urbanization

azone_gq_pop_by_age.csv

azone_hh_pop_by_age.csv

Age55to64

Age65Plus

Apply Preview

Revert Preview

Undo

Redo

Apply Preview changes only this temporary working copy. Save File Changes makes the preview and file note part of Review and the next run.

Saved change from baseline

Unsaved preview

Geo	Year	Age0to14	Age15to19	Age20to29	Age30to54	Age55to64	Age65Plus
Albemarle County	2024	18444	10279	14208	34449	14070	23469
Charlottesville City	2024	6458	2804	11254	14447	4356	6118
Albemarle County	2045	22439	11238	15472	42567	18521	28573
Charlottesville City	2045	26443	3373	13481	19636	6307	8193

Notes

Track assumptions and why this file changes.

32045 only: increase Age0to14, Age15to19, Age20to29, Age30to54, Age55to64, Age65Plus by 10% for Charlottesville City. Whole-number counts are rounded; 2024 is unchanged.

Figure 16. Urbanization changes the Charlottesville City 2045 population record while preserving 2024.

1 Urbanization scenario and file. 2 Edited 2045 values. 3 Note confirms city-only scope and unchanged 2024.

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18. Save the Charlottesville City employment overlay with TotEmp, RetEmp, and SvcEmp increased by 10%.

azone_gq_pop_by_age.csv

azone_hh_pop_by_age.csv

bzone_dwelling_units.csv

bzone_employment.csv

+ New File

Geo

Year

TotEmp

RetEmp

SvcEmp

510030101001

2024

177

0

52

510030102011

2024

543

4

58

510030102012

2024

999

62

624

510030102013

2024

87

0

17

510030102021

2024

115

4

58

510030102022

2024

273

0

109

510030103011

2024

2962

381

1406

510030103021

2024

119

0

74

510030103031

2024

539

107

204

510030103032

2024

156

21

70

510030103033

2024

126

2

66

510030104011

2024

173

0

93

510030104013

2024

985

70

236

510030104021

2024

653

23

49

510030104022

2024

58

0

46

Notes

Track assumptions and why this file changes.

2045 only: increase TotEmp, RetEmp, SvcEmp by 10% for Charlottesville City. Whole-number counts are rounded; 2024 is unchanged.

Figure 17. The employment note completes the city-only Urbanization specification.

7. Review and validate

19. Open Review. Confirm the baseline and all three scenarios are ready to run.

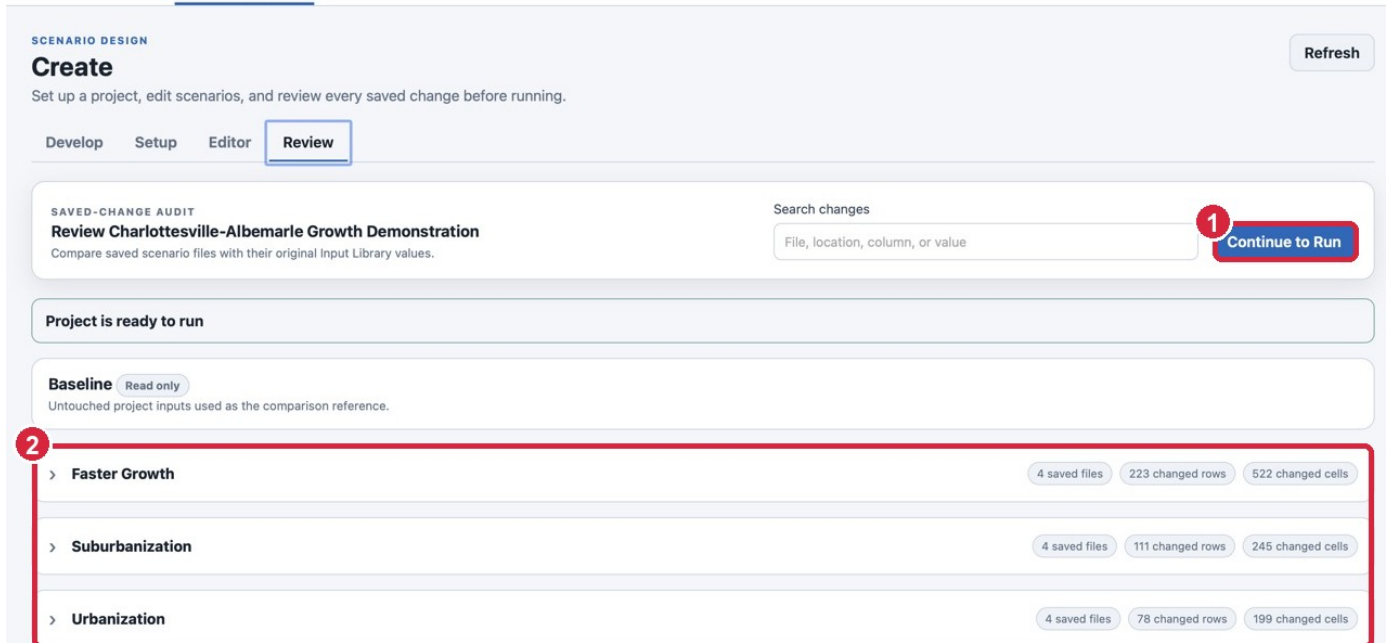


Figure 18. The project-level review shows every scenario ready for execution.

1 Continue to Run is enabled. 2 Each scenario reports four saved files and validation status.

20. Expand each scenario and reconcile saved files, rows, cells, notes, and validation messages.

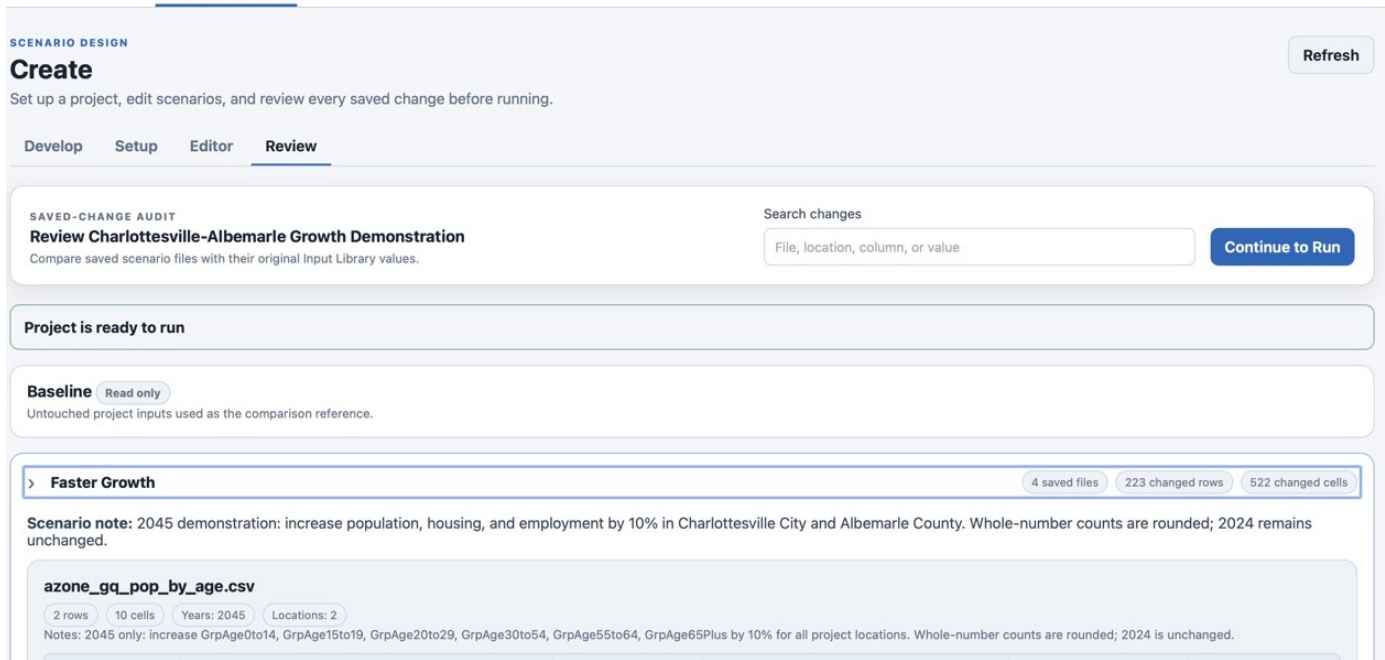


Figure 19. Faster Growth review: four files, 223 changed rows, and 522 changed cells.

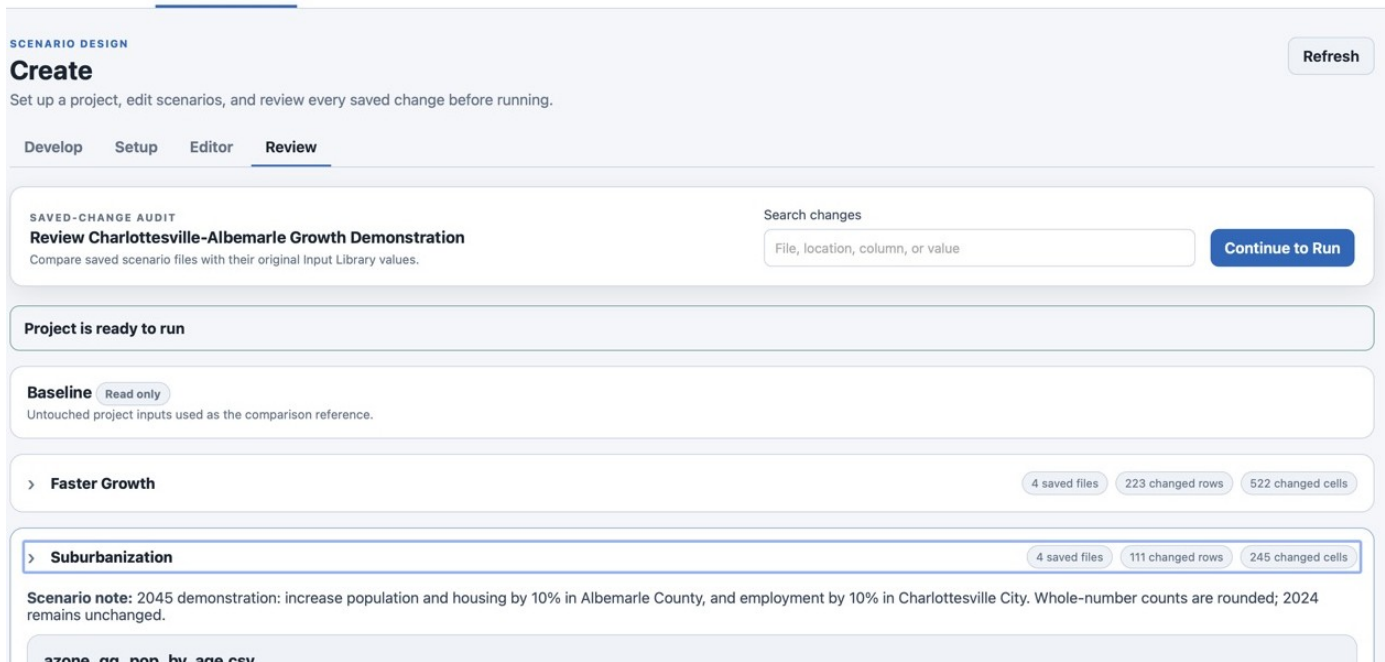


Figure 20. Suburbanization review: four files, 111 changed rows, and 245 changed cells.

Validation record

Scenario	Files	Changed rows	Changed cells	Result
Faster Growth	4	223	522	Ready
Suburbanization	4	111	245	Ready
Urbanization	4	78	199	Ready
		VALIDATION GATE Do not run until every scenario has the correct four files, fields, geography, year 2045, Increase by %, value 10, explanatory note, and ready status.		

8. Run baseline and scenarios

macOS: parallel mode

21. Select baseline and all three scenarios, choose parallel execution, and set two active jobs.
22. Start the batch. Verify that two jobs run while the remaining jobs wait in the queue.

VISIONEVAL RUNTIME

Run

Execute project scenarios through the configured runtime and follow R output live.

Docker **Running** Runtime **Ready** Version **VE-40-RC6**

Docker and the verified VisionEval runtime are ready. Containers are created temporarily for each run.

BATCH SETUP

Select runs
Nothing selected — choose each run explicitly

Project: Charlottesville-Albemarle Growth Demonstration Run selected

1 ☐ **Baseline**
Fresh untouched model inputs

☐ **Faster Growth**
4 edited input files

☐ **Suburbanization**
4 edited input files

☐ **Urbanization**
4 edited input files

RUN HISTORY

Jobs Hide run history Refresh

2 **Baseline**
Charlottesville-Albemarle Growth Demonstration · 8/17/2026, 8:20:21 PM
VisionEval is running

Faster Growth
Charlottesville-Albemarle Growth Demonstration · 8/17/2026, 8:20:21 PM
VisionEval is running

LIVE OUTPUT

Baseline · running Stop Run Stop All Runs

3 **Suburbanization · waiting**

2026-08-18 00:20:23 : Initializing Model. This may take some time...

2026-08-18 00:20:27 : Initializing VEPowertrainsAndFuels

2026-08-18 00:20:27 : Initializing VETravelPerformance

Name: VERSPM

Scenario: Base

Description: Region Builder generated Virginia region for Charlottesville-Albemarle Metropolitan Planning Organ

Timestamp: 2026-08-18 00:20:23.528582

2026-08-18 00:20:36 : Model successfully initialized.

Figure 21. Two jobs are active and the remaining scenarios are queued on macOS.

- 1** All four runs belong to one batch. **2** Two active job slots. **3** Unstarted jobs remain queued until a slot opens.

VISIONEVAL RUNTIME

Run

Execute project scenarios through the configured runtime and follow R output live.

Docker **Running** Runtime **Ready** Version **VE-40-RC6**

Docker and the verified VisionEval runtime are ready. Containers are created temporarily for each run.

BATCH SETUP

Select runs
Nothing selected — choose each run explicitly

Project: Charlottesville-Albemarle Growth Demonstration Run selected

☐ **Baseline**
Fresh untouched model inputs

☐ **Faster Growth**
4 edited input files

☐ **Suburbanization**
4 edited input files

☐ **Urbanization**
4 edited input files

RUN HISTORY

Jobs Hide run history Refresh

1 **Suburbanization**
Charlottesville-Albemarle Growth Demonstration · 8/17/2026, 8:20:21 PM
VisionEval is running

Urbanization
Charlottesville-Albemarle Growth Demonstration · 8/17/2026, 8:20:21 PM
VisionEval is running

HISTORY

Faster Growth
Charlottesville-Albemarle Growth Demonstration · 8/17/2026, 8:20:21 PM
VisionEval is running

LIVE OUTPUT

Suburbanization · running Stop Run Stop All Runs

2 **Urbanization · running**

2026-08-18 00:23:37 : Finish module 'VELandUse::AssignDemandManagement' for year '2024'.

2026-08-18 00:23:38 : Start module 'VELandUse::AssignCarSvcAvailability' for year '2024'.

2026-08-18 00:23:38 : Finish module 'VELandUse::AssignCarSvcAvailability' for year '2024'.

2026-08-18 00:23:38 : Start module 'VETransportSupply::AssignTransitService' for year '2024'.

2026-08-18 00:23:38 : Start module 'VETransportSupply::AssignRoadMiles' for year '2024'.

2026-08-18 00:23:38 : Start module 'VETransportSupply::AssignRoadMiles' for year '2024'.

2026-08-18 00:23:38 : Start module 'VEHouseholdVehicles::AssignDrivers' for year '2024'.

2026-08-18 00:23:42 : Finish module 'VEHouseholdVehicles::AssignVehicleOwnership' for year '2024'.

2026-08-18 00:23:42 : Start module 'VEHouseholdVehicles::AssignVehicleOwnership' for year '2024'.

2026-08-18 00:23:46 : Finish module 'VEHouseholdVehicles::AssignVehicleType' for year '2024'.

2026-08-18 00:23:52 : Finish module 'VEHouseholdVehicles::AssignVehicleType' for year '2024'.

2026-08-18 00:23:53 : Start module 'VEHouseholdVehicles::CreateVehicleTable' for year '2024'.

2026-08-18 00:23:53 : Finish module 'VEHouseholdVehicles::CreateVehicleTable' for year '2024'.

2026-08-18 00:23:54 : Start module 'VEHouseholdVehicles::AssignVehicleAge' for year '2024'.

Figure 22. When a slot opens, the next queued scenario starts automatically.

Windows: queued execution only

WINDOWS LIMITATION Parallel runs are unavailable on Windows. Select queued/single-run execution and allow each run to finish before the next begins. Do not open multiple Workbench instances to imitate parallel mode.

23. Wait until all four jobs report Succeeded, then verify and register each datastore.

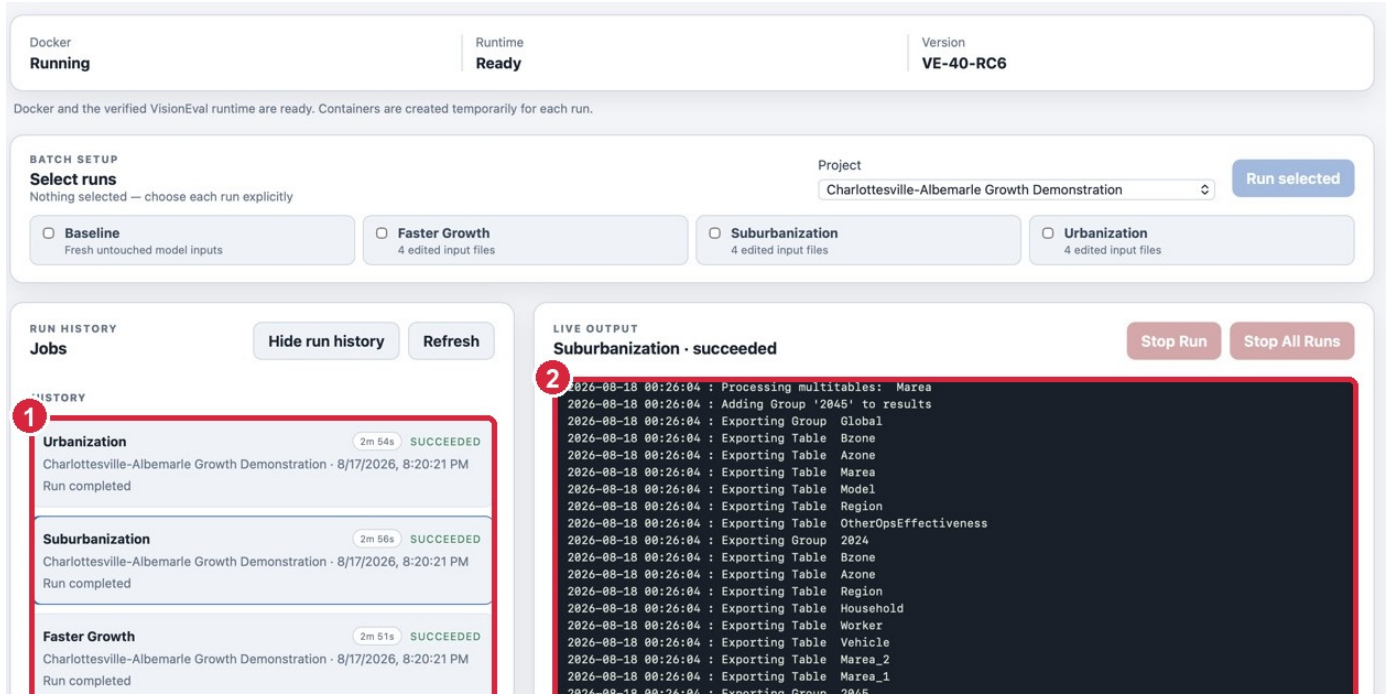


Figure 23. Baseline and all three scenario runs succeeded and their datastores were verified.

1 Succeeded and verified status. 2 Job consoles provide an audit trail for each run.

Completed datastore record

Result	Run status	Datastore status
Baseline	Succeeded	Verified / registered
Faster Growth	Succeeded	Verified / registered
Suburbanization	Succeeded	Verified / registered
Urbanization	Succeeded	Verified / registered
		ACTUAL RESULTS ONLY The comparison figures in this guide come from these completed datastores. No comparison output was staged or mocked.

9. Compare observed outputs

9.1 Load Baseline vs. Suburbanization and Urbanization

24. Open Compare → Load Data. Choose Baseline as the reference, Suburbanization as Comparison 1, and Urbanization as Comparison 2.

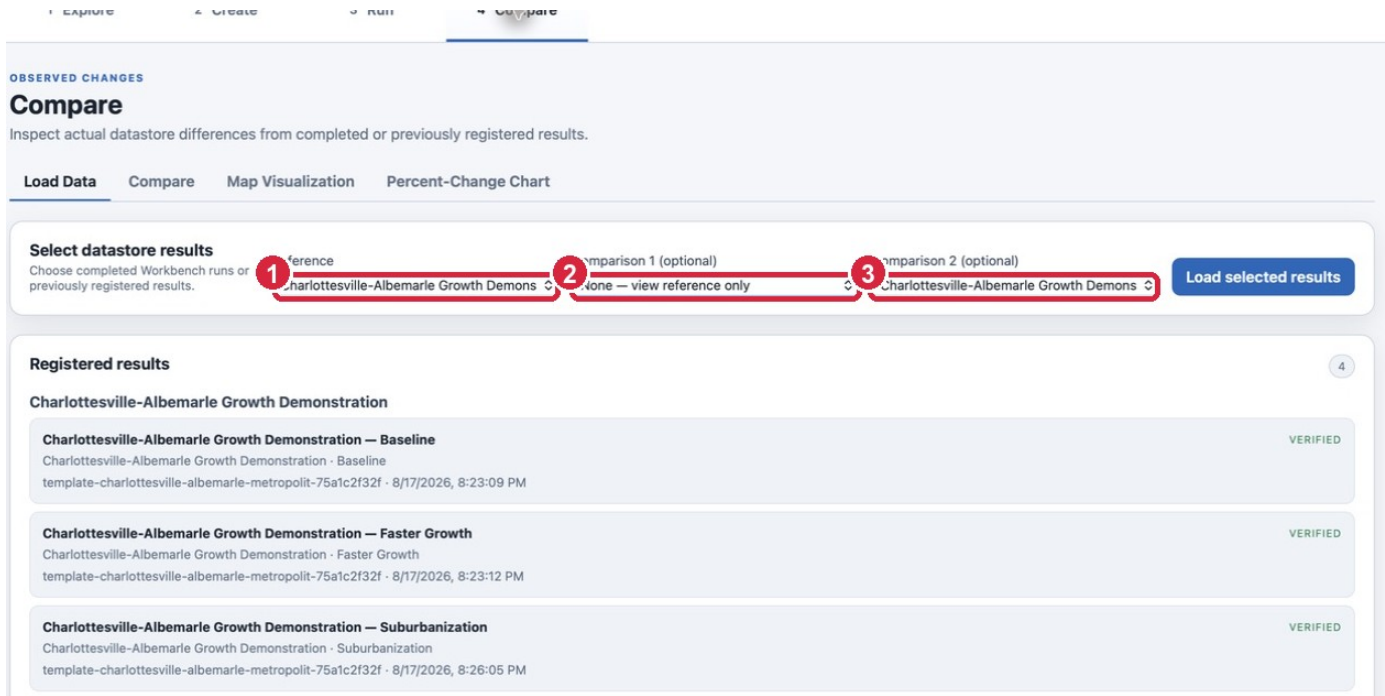


Figure 24. The first comparison configuration keeps one baseline and loads both location-pattern alternatives.

1 Baseline reference. 2 Suburbanization comparison. 3 Urbanization comparison.

25. Load the selected results and wait for the comparable-output inventory to finish.

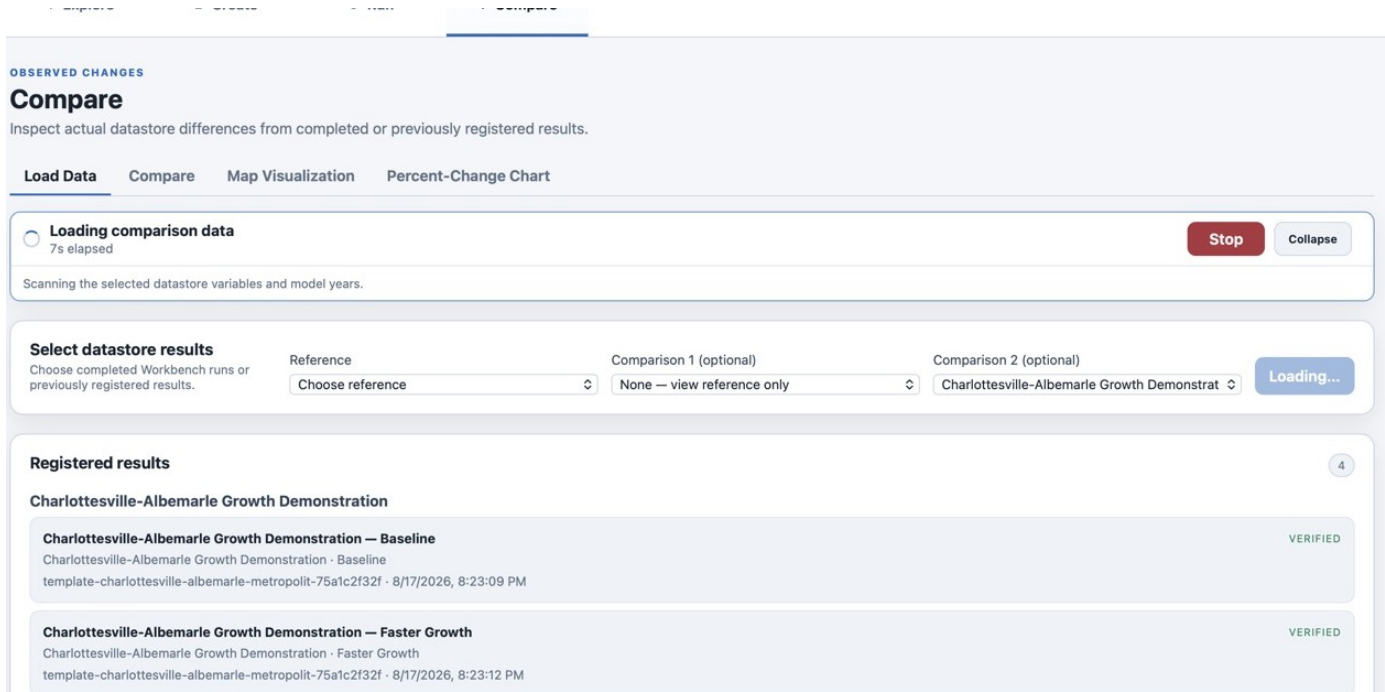


Figure 25. All three completed datastores are registered and available for comparison.

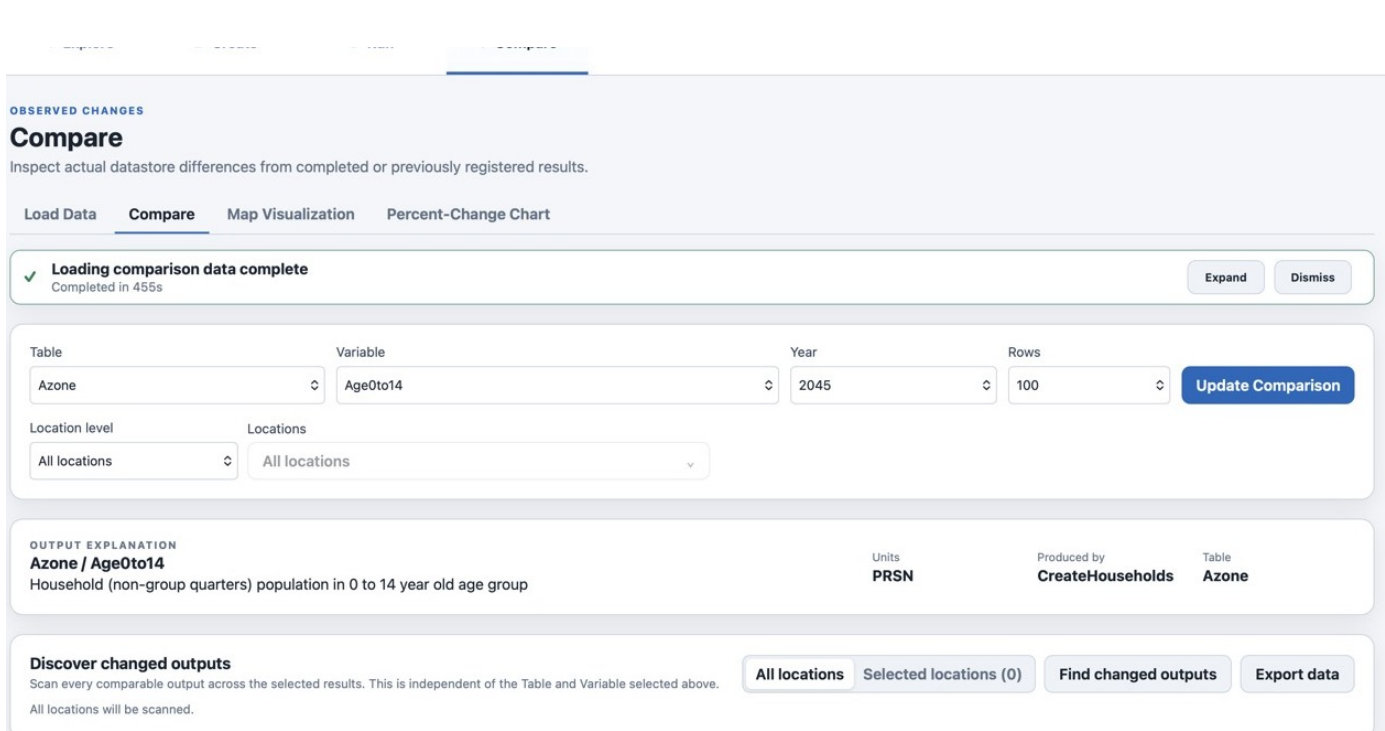


Figure 26. The detailed comparison view is active for a selected 2045 output.

9.2 Discover outputs that actually changed

26. Choose Find changed outputs. Scan all locations first; narrow geography only after confirming model-wide changes.

OBSERVED CHANGES

Compare

Inspect actual datastore differences from completed or previously registered results.

Load Data

Compare

Map Visualization

Percent-Change Chart

✓

Finding changed outputs complete

Completed in 9s

Expand

Dismiss

Table

Variable

Year

Rows

Azone

Age0to14

2045

100

Update Comparison

Location level

Locations

All locations

All locations

OUTPUT EXPLANATION

Azone / Age0to14

Household (non-group quarters) population in 0 to 14 year old age group

Units

PRSN

Produced by

CreateHouseholds

Table

Azone

Discover changed outputs

Scan every comparable output across the selected results. This is independent of the Table and Variable selected above.

All locations

Selected locations (0)

Find changed outputs

Export data

All locations will be scanned.

Figure 27. Changed-output discovery scans the completed datastores rather than relying on expected variables.

Discover changed outputs

Scan every comparable output across the selected results. This is independent of the Table and Variable selected above.

All locations

Selected locations (0)

Find changed outputs

Export data

All locations will be scanned.

1

203 changed variables

Scanned 401; skipped 1 unsafe or unavailable variables.

2

Output ---	Charlottesville-Albemarle Growth Demonstration — Suburbanization change % ---
Vehicle / MPGe	-0.27%
Vehicle / MPG	-0.83%
Vehicle / FuelCO2ePM	0.45%
Vehicle / OwnCost	2.14%
Vehicle / OwnCostPerMile	-22.95%
Vehicle / DvmtProp	-2.06%
Vehicle / GPM	0.45%
Vehicle / Age	0.17%
Vehicle / InsCost	1.72%
Vehicle / MPKWH	-3.67%
Vehicle / ElecCO2ePM	-3.17%
Worker / DistanceToWork	-11.13%
Vehicle / KWHPM	-3.63%
Vehicle / Powertrain	Not available

Figure 28. The completed scan found 203 changed variables among 401 scanned; one unsafe or unavailable variable was skipped.

- 1
- Observed changed-output count.
- 2
- Select a stable output from the returned list.

9.3 Map a stable 2045 output

27. Open Map Visualization. Use 2045, table Bzone, variable Pop, and County/locality geography.

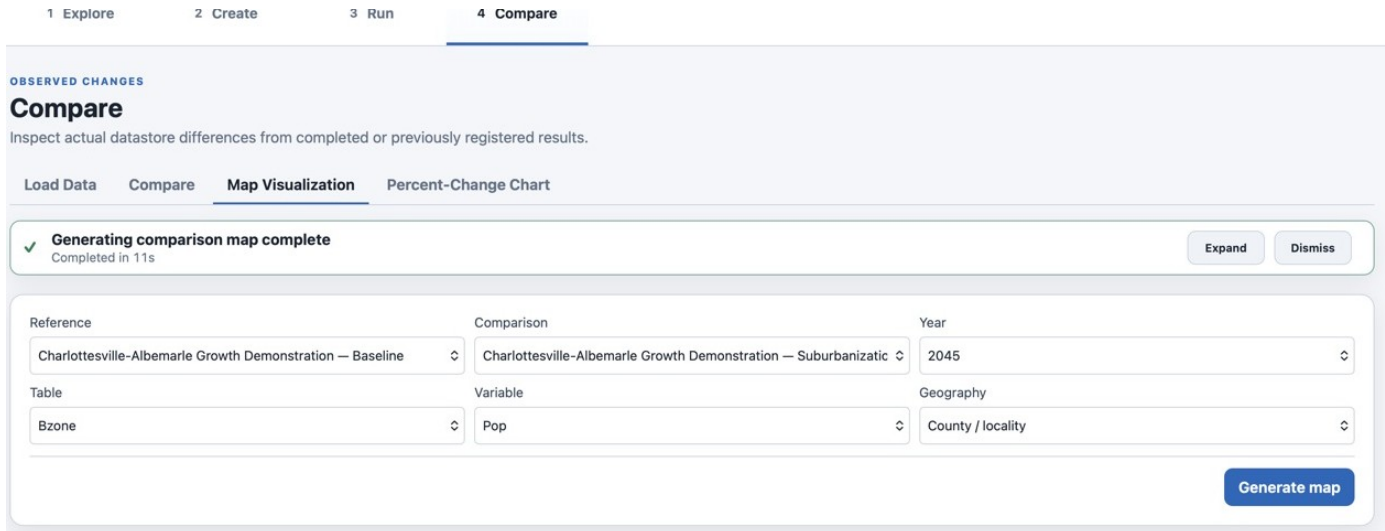


Figure 29. The map configuration uses a stable Bzone population output and county/locality geography.

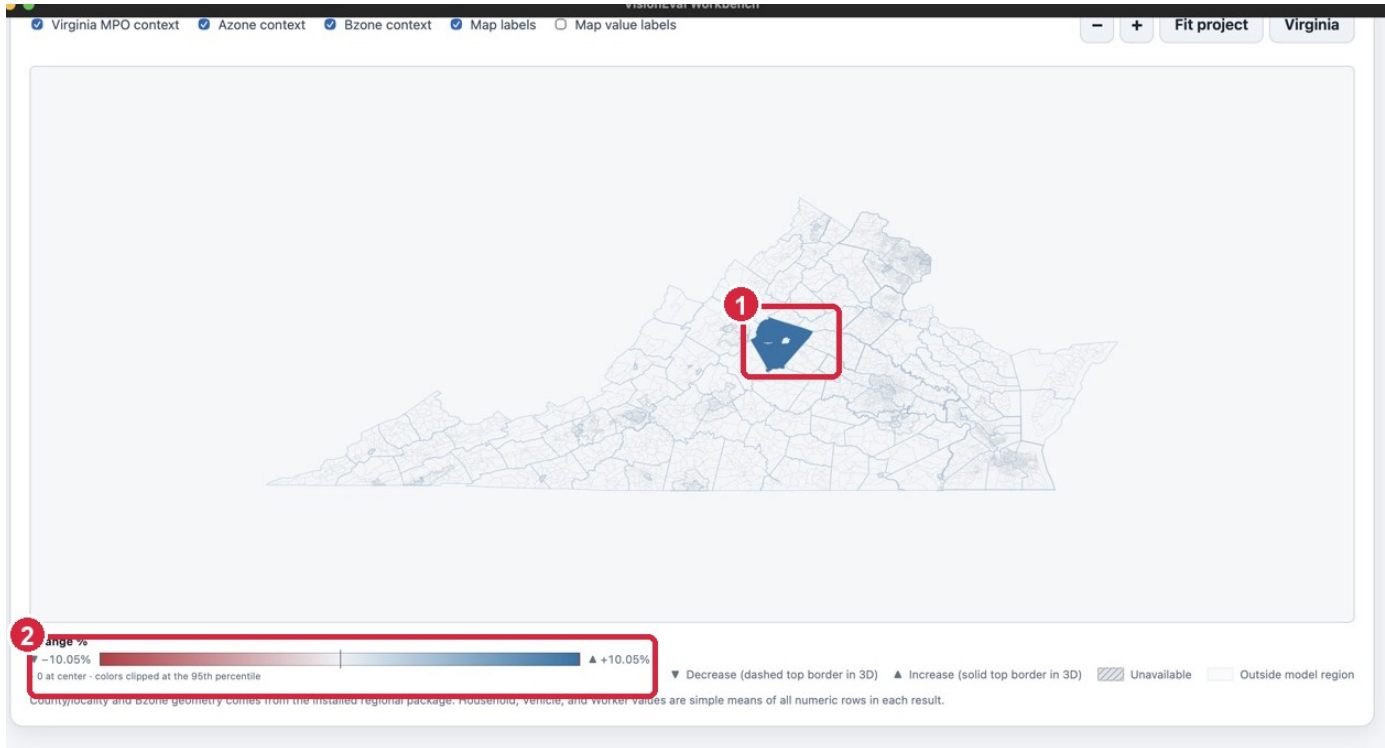


Figure 30. Observed Suburbanization map: Charlottesville City is approximately +10.05% while Albemarle County is near zero for this selected output.

- 1 Charlottesville City carries the visible increase. 2 The legend reports observed percent change, not the input assumption.

OUTPUT SUBSTITUTION Bzone/Pop, Bzone/TotEmp, Bzone/NumHh, and Bzone/NumWkr were available and stable. A direct Bzone/Dvmt entry was not available in the map selector, so the tutorial uses Bzone/Pop as the first stable equivalent confirmed by changed-output discovery.

9.4 Generate a percent-change chart

28. Open Percent-Change Chart. Keep Baseline and Suburbanization, select 2045, and search for Bzone/Pop.

OBSERVED CHANGES

Compare

Inspect actual datastore differences from completed or previously registered results.

Load Data

Compare

Map Visualization

Percent-Change Chart

Generating chart

2m 34s elapsed

Stop

Collapse

Scanning selected numeric variables.

Reference

Comparison

Year

Charlottesville-Albemarle Growth Demonstration — Baseline

Charlottesville-Albemarle Growth Demonstration — Suburbanizatic

2045

Location level

Locations

All locations

All locations

Variables

All numeric outputs

Search variables

Bzone/Pop

Select visible

Clear

Choose outputs to chart. With none selected, all comparable numeric outputs are generated.

Figure 31. Workbench generates the percent-change chart from the same actual registered datastores.

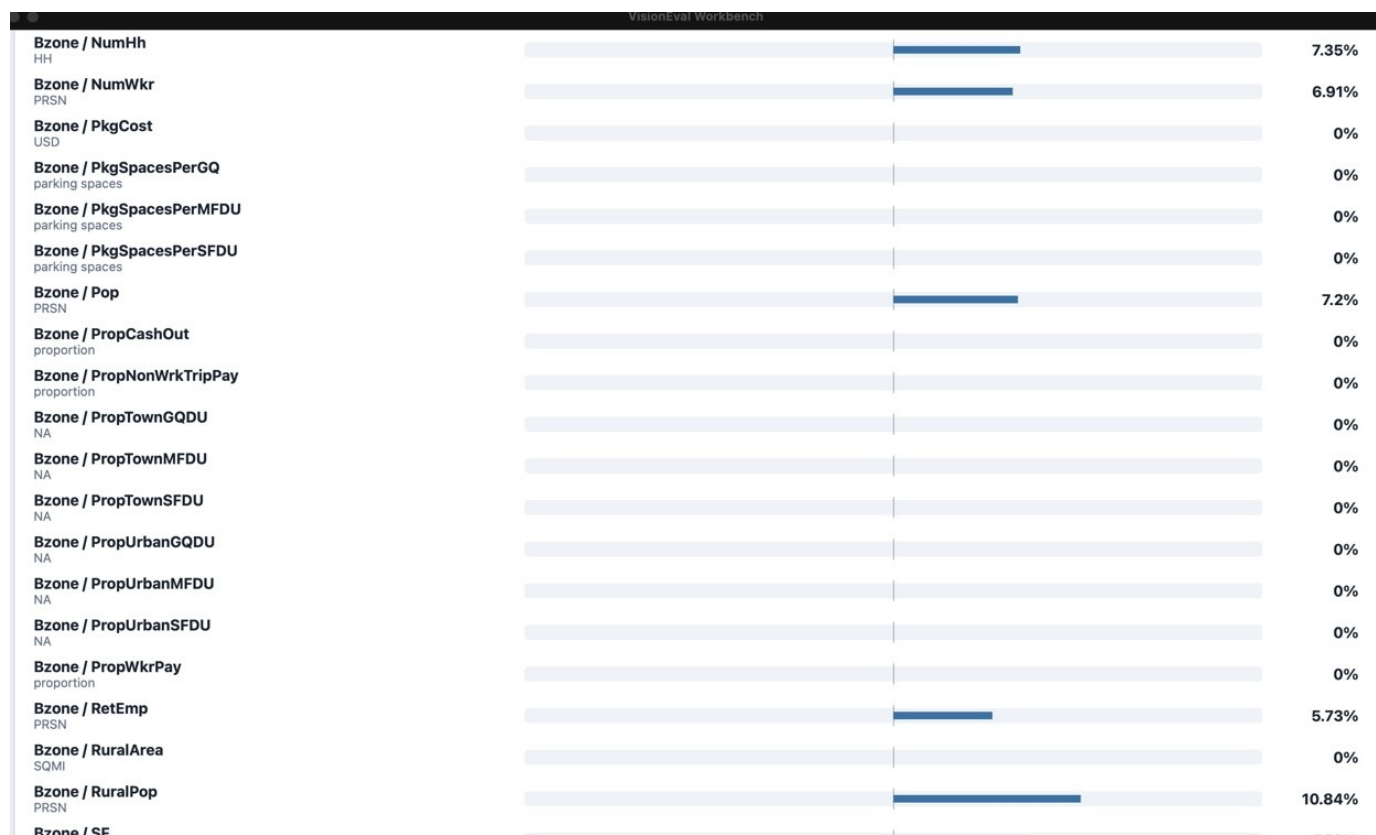


Figure 32. The completed chart reports observed aggregate changes, including Bzone/Pop at 7.2%, Bzone/NumHh at 7.35%, and Bzone/NumWkr at 6.91%.

9.5 Load Baseline vs. Faster Growth

29. Return to Load Data. Keep Baseline as reference, choose Faster Growth as Comparison 1, and clear Comparison 2.

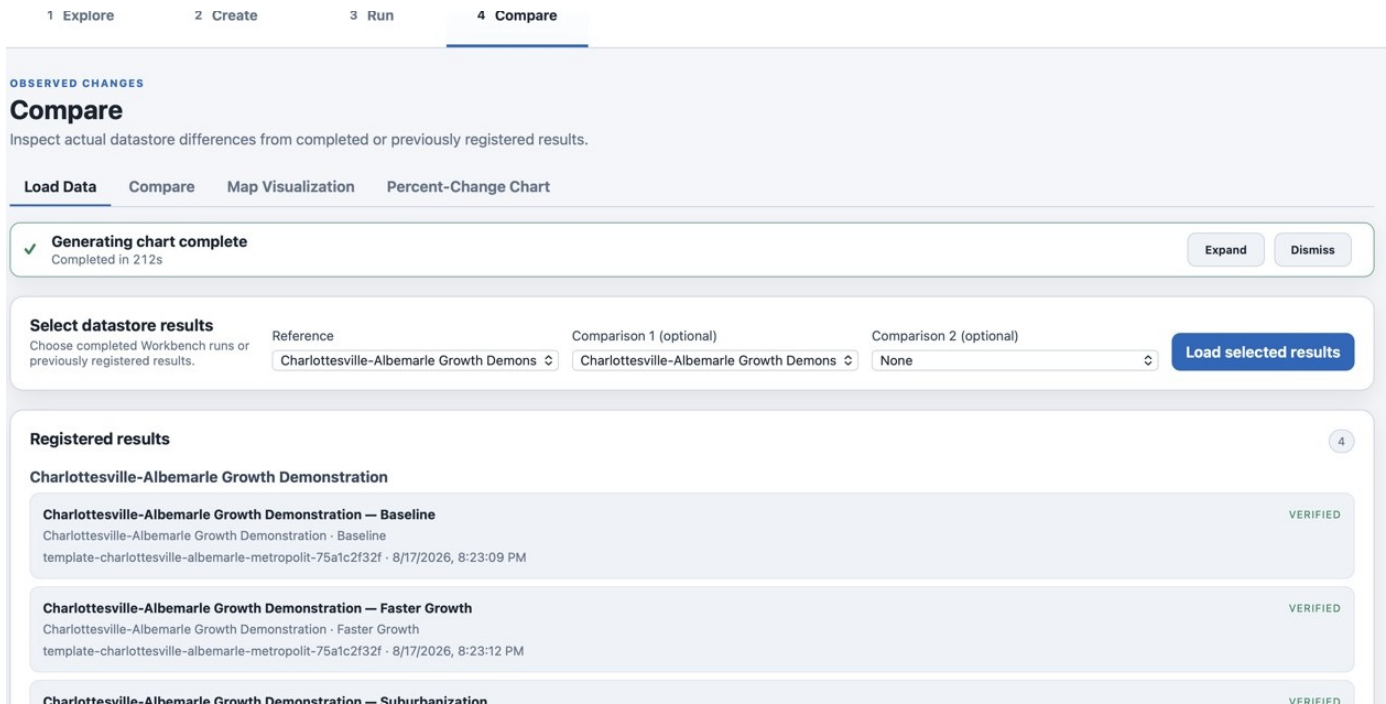


Figure 33. The second configuration isolates Faster Growth against the same verified baseline.

This second configuration answers a different question: the two location-pattern scenarios show redistribution, while Faster Growth shows a region-wide increase. Reuse changed-output discovery before selecting maps or charts; do not assume the same outputs will be the most informative.

INTERPRETATION RULE Input changes are assumptions; output changes are model results. Report the observed output percentage shown by Workbench, not the +10% input value, unless the output truly changes by exactly 10%.

10. Export results

30. **Set the view.** In Compare, choose the comparison, table, variable, year, and geography you want to export.
31. **Choose the export type.**
 - DATA** Choose Export data, then select CSV or Excel.
 - VISUAL** Open Map Visualization or Percent-Change Chart, confirm the displayed comparison and filters, then use the visual export control.
32. **Check the saved file.** Open the export and confirm it matches the comparison, year, geography, labels, and values shown in Workbench.

11. Settings and preferences

Settings controls the active workspace and the defaults Workbench uses for future work. Open it with the gear button. On macOS, you can also press **⌘,**; on Windows, press **Ctrl+,**. Settings remain open after saving so you can confirm the status message.

11.1 Workspace and assets

33. **Open Settings.** Confirm the active workspace before changing projects, assets, runs, or results.
34. **Review workspace and asset controls.** Switch or move the workspace only when no job is active or waiting. Asset defaults apply to new projects only; existing projects remain pinned to their recorded template and InputLibrary.

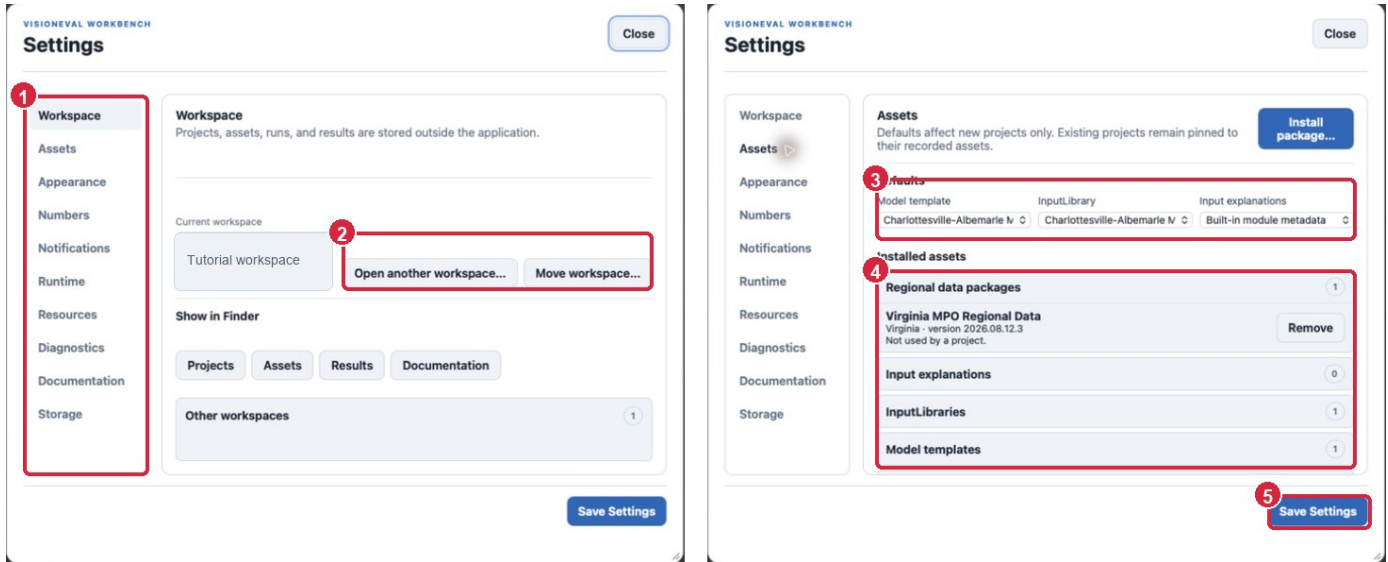


Figure 34. Workspace and Assets identify where work is stored and which assets new projects will use.

- 1 Settings categories. 2 Open or move the workspace. 3 Defaults used by new projects. 4 Installed and removed asset groups. 5 Save the settings change.

Settings at a glance

Settings page	What it controls
Workspace	View, switch, or move the workspace and reveal Projects, Assets, Results, or Documentation.
Assets	Install packages, choose defaults for new projects, and inspect or remove eligible assets.
Appearance	Choose System, Light, or Dark and set accessible decrease, neutral, and increase colors.
Numbers	Set default decimal places and optional calculation, output, and percentage overrides.
Notifications	Enable desktop alerts, choose the successful-work delay, and send a test notification.
Runtime	Install, verify, or repair the platform-specific VisionEval runtime.
Resources	Choose the default run mode and optional memory behavior; at most two jobs run concurrently.
Diagnostics	Build a failed-run support bundle and optionally include the datastore or comparison cache.
Documentation	Read the managed Workbench User Guide inside the application.
Storage	Control retained full CSV exports, review usage, and clear or rebuild the disposable comparison cache.

11.2 Appearance and number display

- 35. **Choose the display preferences.** Select System, Light, or Dark; then set accessible decrease, neutral, and increase colors for comparison views.
- 36. **Choose number precision.** Set the default decimal places and expand Advanced overrides only when calculation or display contexts need different precision.

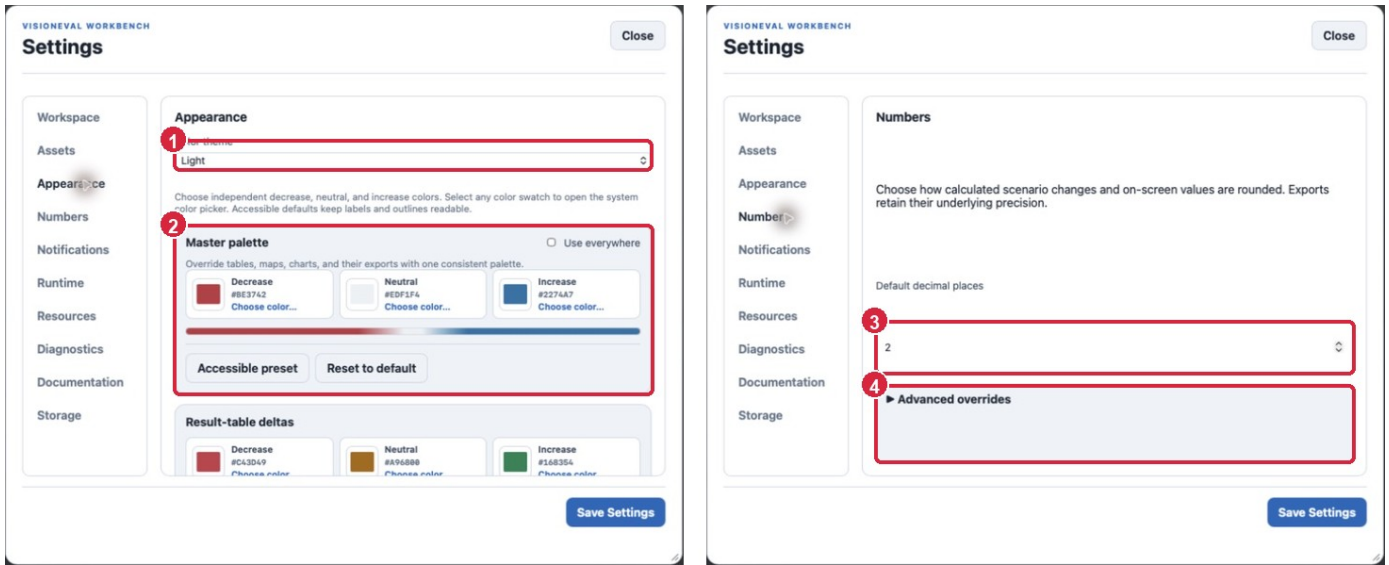


Figure 35. Appearance controls themes and comparison colors; Numbers controls calculation and display precision.

- 1 System, Light, or Dark theme. 2 Decrease, neutral, and increase palette. 3 Default decimal places. 4 Optional precision overrides.

PRECISION Number-display settings do not change authoritative raw values or the underlying precision retained by data exports.

11.3 Notifications

37. **Configure background-work alerts.** Enable desktop notifications, choose the minimum duration before successful work triggers an alert, and use Send Test Notification to confirm delivery.

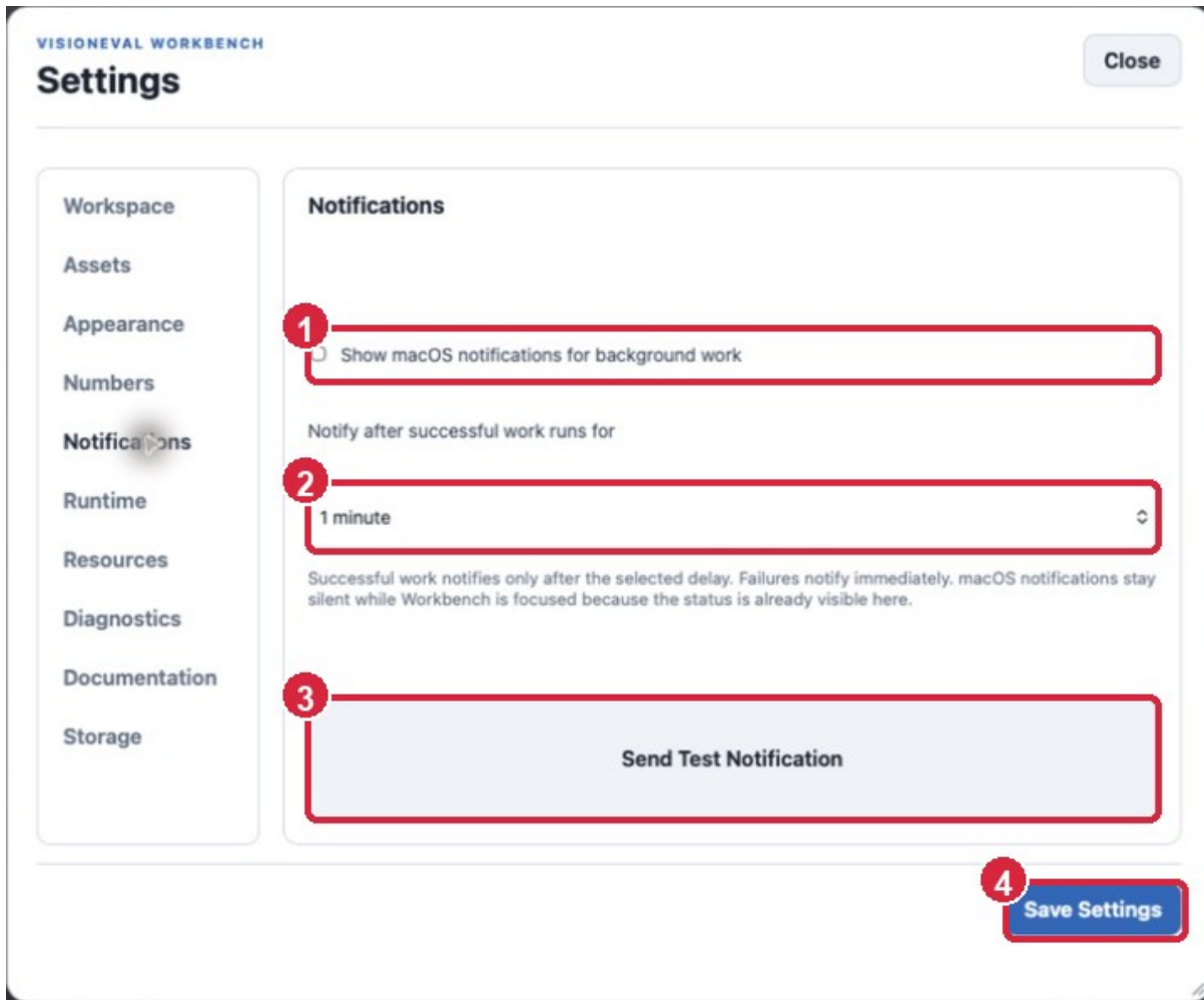


Figure 36. Notifications can report completed or failed background work when Workbench is not focused.

- 1 Enable desktop notifications. 2 Choose the successful-work delay. 3 Send a test notification. 4 Save the preference.

Failures notify immediately when notifications are enabled. Successful work observes the selected delay, and notifications remain silent while Workbench is focused because status is already visible. Operating-system notification permissions still apply.

11.4 Runtime, resources, diagnostics, and storage

38. **Check Runtime and Resources.** Use Runtime to install, verify, or repair the platform-specific VisionEval connection. Resources controls the default run mode and optional memory behavior; Workbench allows at most two concurrent VisionEval jobs.
39. **Create diagnostics only when needed.** Select a failed run, keep the optional datastore and cache additions off unless they are relevant, then export the bundle for technical review.
40. **Manage storage safely.** Review usage, decide whether full CSV exports should be retained, and clear or rebuild the disposable comparison cache without removing registered results.

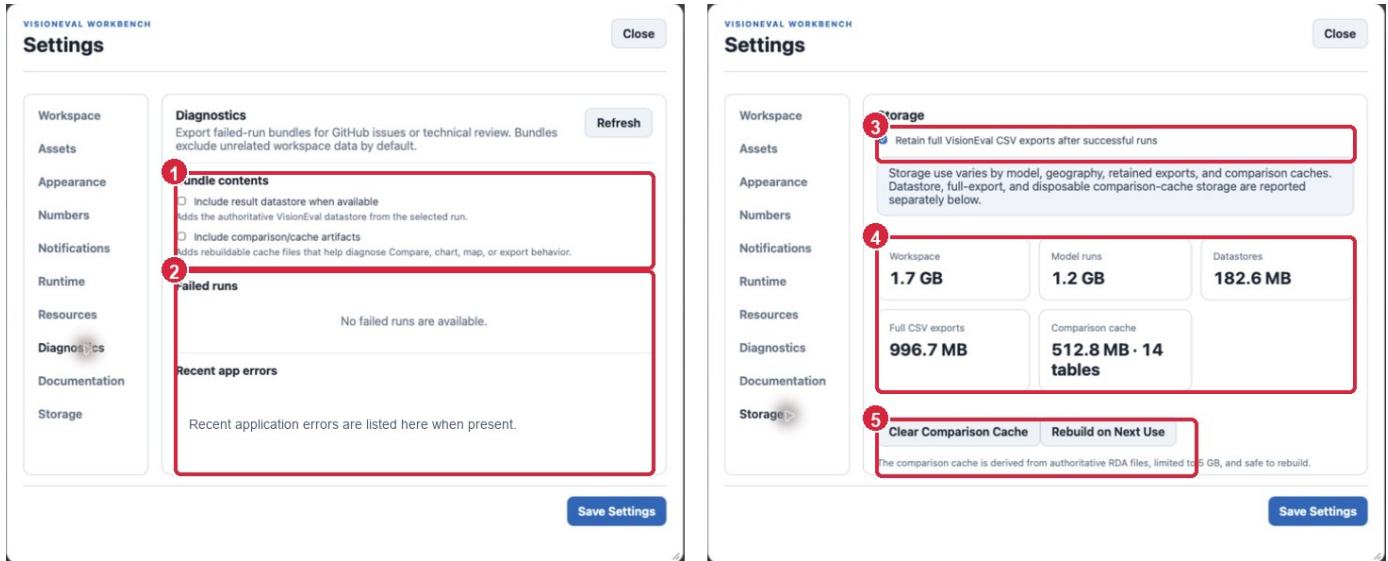


Figure 37. Diagnostics packages troubleshooting evidence; Storage reports retained and rebuildable data separately.

- 1 Optional diagnostic-bundle additions. 2 Failed runs and recent application errors. 3 Retain full CSV exports. 4 Storage usage by category. 5 Clear or rebuild the comparison cache.
41. **Save and verify.** Select Save Settings and read the footer status. Documentation can be opened from its Settings page without leaving Workbench.

12. Appendix: Project management and region reuse

These tasks are separate from the scenario exercise, but they are useful when cleaning up completed work or starting another study for a region that has already been built.

12.1 Remove, restore, or permanently delete a project

42. **Open Create → Setup.** Find the project under Saved work and select Remove.
43. **Confirm the archive action.** Remove hides the project and its jobs and results from normal Create, Run, and Compare lists, but keeps it recoverable for 30 days.

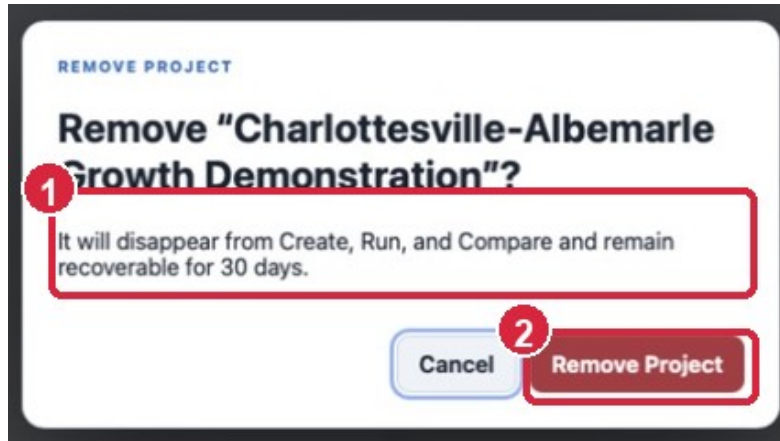


Figure 38. Remove Project clearly states that the project remains recoverable for 30 days.

- 1 Archive effect and recovery period. 2 Confirm the recoverable removal.
44. **Open Archived Projects.** Choose Restore to return the project to normal lists. Choose Delete now only when permanent removal is intended.

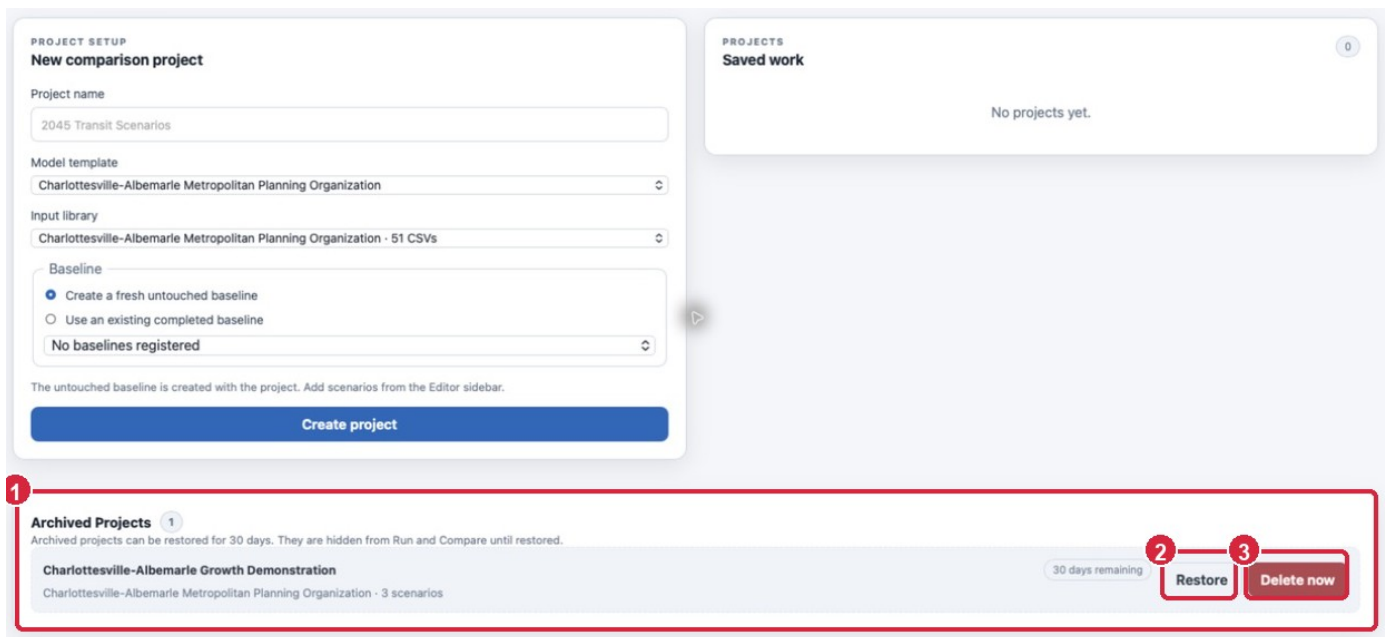


Figure 39. Archived Projects shows the recovery period and the available Restore and Delete now actions.

- 1 Archived project and days remaining. 2 Restore the project. 3 Permanently delete eligible data.

PROJECT SAFETY A project with an active or waiting job cannot be archived. Use Workbench—not Finder—to remove managed projects. Delete now is permanent.

12.2 Start another project for the same MPO

Region Builder creates a reusable model-template and InputLibrary pair. A new project for the same MPO can select that pair directly; the region does not need to be built again.

45. **Return to Create → Setup.** Enter a distinct project name so the new study is easy to identify.
46. **Reuse the regional assets.** Select the existing Charlottesville–Albemarle model template and InputLibrary.
47. **Choose the baseline.** Create a fresh untouched baseline for an independent project, or use an eligible completed baseline only when that dependency is intentional.
48. **Create the project.** Workbench creates a separate project with its own baseline, scenarios, notes, jobs, and results while reusing the existing regional asset pair.

Figure 40. Create Setup reuses the existing Charlottesville–Albemarle assets for a separate project.

- 1 New project name. 2 Existing regional template and InputLibrary. 3 Independent or intentionally shared baseline. 4 Create the separate project.

WHEN TO REBUILD Return to Develop and rebuild the region only when the generated assets were removed, replaced by an updated package, or intentionally regenerated. A different project name or scenario purpose alone does not require a rebuild.

Project home: [VisionEval Workbench on GitHub](#)

Problems or ideas for improvement? [Open a GitHub Issue](#) to report a problem or suggest a change. For run failures, include the diagnostics bundle from Settings → Diagnostics.

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